

### 5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

#### Sample Data:

| PassengerId | Survived | Pclass | Name                                                | Sex    | Age | SibSp | Parch | Ticket           | Fare    | Cabin | Embarked |
|-------------|----------|--------|-----------------------------------------------------|--------|-----|-------|-------|------------------|---------|-------|----------|
| 1           | 0        | 3      | Braund, Mr. Owen Harris                             | male   | 22  | 1     | 0     | A/5 21171        | 7.25    |       | S        |
| 2           | 1        | 1      | Cumings, Mrs. John Bradley (Florence Briggs Thayer) | female | 38  | 1     | 0     | PC 17599         | 71.2833 | C85   | C        |
| 3           | 1        | 3      | Heikkinen, Miss. Laina                              | female | 26  | 0     | 0     | STON/O2. 3101282 | 7.925   |       | S        |
| 4           | 1        | 1      | Futrelle, Mrs. Jacques Heath (Lily May Peel)        | female | 35  | 1     | 0     | 113803           | 53.1    | C123  | S        |
| 5           | 0        | 3      | Allen, Mr. William Henry                            | male   | 35  | 0     | 0     | 373450           | 8.05    |       | S        |
| 6           | 0        | 3      | Noran, Mr. James                                    | male   |     | 0     | 0     | 330877           | 8.4583  |       | Q        |
| 7           | 0        | 1      | McCarthy, Mr. Timothy J                             | male   | 54  | 0     | 0     | 17463            | 51.8625 | E46   | S        |
| 8           | 0        | 3      | Palsson, Master. Gosta Leonard                      | male   | 2   | 3     | 1     | 349909           | 21.075  |       | S        |
| 9           | 1        | 3      | Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)   | female | 27  | 0     | 2     | 347742           | 11.1333 |       | S        |
| 10          | 1        | 2      | Nasser, Mrs. Nicholas (Adele Achem)                 | female | 14  | 1     | 0     | 237736           | 30.0708 |       | C        |

AgeFareS...SUBMIT

```
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15 """
16 import pandas as pd
17 import matplotlib.pyplot as plt
18
19 # Load the Titanic dataset
20 data = pd.read_csv('Titanic-Dataset.csv')
21
22 # Data Cleaning
23 data['Age'].fillna(data['Age'].median(), inplace=True)
24 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
25 data.drop('Cabin', axis=1, inplace=True)
26
27 # Convert categorical features to numeric
28 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
29 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
30
31 # Write your code here for Scatter Plot for Age vs. Fare by Survived
32 colors = data['Survived'].map({0: 'red', 1: 'blue'})
33
34 plt.scatter(data['Age'], data['Fare'], c=colors)
35 plt.title('Age vs. Fare by Survival')
36 plt.xlabel('Age')
37 plt.ylabel('Fare')
38 plt.show()
39
40
41
42
```

TerminalTest cases

Debugger

## 5.2.10. Scatter Plot for Age vs. Fare

01:27

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

## Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

**Note:** Refer to the visible test case for better reference.

AgeFareS...

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```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Fare by Pclass
17 # Write your code here for Box Plot for Fare by Pclass
18 # Write your code here for Scatter Plot for Age vs. Fare
19 plt.scatter(data['Age'], data['Fare'])
20 plt.title('Age vs. Fare')
21 plt.xlabel('Age')
22 plt.ylabel('Fare')
23 plt.show()
24
25
26
27
```

Terminal Test cases

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### 5.2.9. Box Plot for Fare by Pclass

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Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

#### Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

#### BoxPlotF...

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```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Fare by Pclass
17 # Write your code here for Box Plot for Fare by Pclass
18 # Write your code here for Box Plot for Fare by Pclass
19
20 # 1. Use the Pclass column to group the data for the boxplot.
21 data.boxplot(column='Fare', by='Pclass')
22
23 # 2. Set the title of the plot to "Fare by Pclass".
24 plt.title('Fare by Pclass')
25
26 # 3. Remove the default subtitle.
27 plt.suptitle('')
28
29 # 4. Label the x-axis as 'Pclass' and the y-axis as 'Fare'.
```

Terminal

Test cases

5.2.8. Box Plot for Age by Survived

01:30

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

- 1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
- 2. Set the title of the plot to **"Age by Survival"**.
- 3. Remove the default subtitle with **plt.suptitle("")**.
- 4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

BoxPlotF...

1 import pandas as pd  
2 import matplotlib.pyplot as plt  
3  
4 # Load the Titanic dataset  
5 data = pd.read\_csv('Titanic-Dataset.csv')  
6  
7 # Data Cleaning  
8 data['Age'].fillna(data['Age'].median(), inplace=True)  
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)  
10 data.drop('Cabin', axis=1, inplace=True)  
11  
12 # Convert categorical features to numeric  
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})  
14 data = pd.get\_dummies(data, columns=['Embarked'], drop\_first=True)  
15  
16 # Write your code here for Box Plot for Age by Survived  
17 # Write your code here for Box Plot for Age by Survived  
18 # Write your code here for Box Plot for Age by Survived  
19 data.boxplot(column='Age', by='Survived')  
20 plt.title('Age by Survival')  
21 plt.suptitle('') # Remove the default subtitle  
22 plt.xlabel('Survived')  
23 plt.ylabel('Age')  
24 plt.show()  
25  
26

Terminal Test cases

5.2.7. Box plot for Age Distribution

01:08

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

- 1. Use the **Pclass** column to group the data for the boxplot.
- 2. Set the title of the plot to **"Age by Pclass"**.
- 3. Remove the default subtitle with **plt.suptitle("")**.
- 4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

BoxPlotF... SUBMIT

1 import pandas as pd  
2 import matplotlib.pyplot as plt  
3  
4 # Load the Titanic dataset  
5 data = pd.read\_csv('Titanic-Dataset.csv')  
6  
7 # Data Cleaning  
8 data['Age'].fillna(data['Age'].median(), inplace=True)  
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)  
10 data.drop('Cabin', axis=1, inplace=True)  
11  
12 # Convert categorical features to numeric  
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})  
14 data = pd.get\_dummies(data, columns=['Embarked'], drop\_first=True)  
15  
16 # Write your code here for Box Plot for Age by Pclass  
17 # Write your code here for Box Plot for Age by Pclass  
18  
19 # Create the boxplot  
20 data.boxplot(column='Age', by='Pclass')  
21  
22 # Set the title and labels  
23 plt.title('Age by Pclass')  
24 plt.suptitle('') # Remove the default automatic subtitle  
25 plt.xlabel('Pclass')  
26 plt.ylabel('Age')  
27  
28 # Display the plot  
29 plt.show()  
30

Terminal Test cases

5.2.6. Bar Plot for Survival by Embarked

02:01

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the **Embarked\_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "Survival by Embarked " to the chart.
4. Label the x-axis as 'Embarked' and the y-axis as 'Count'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as 'Survived' and 'Not Survived').

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
```

BarPlotOf...

SUBMIT

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Embarked
17
18 # Grouping data by embarkation point and survival status
19 survival_counts = data.groupby('Embarked_Q')
20 ['Survived'].value_counts().unstack().fillna(0)
21
22 # Plotting the stacked bar chart
23 survival_counts.plot(kind='bar', stacked=True)
24
25 # Adding labels and titles
26 plt.title('Survival by Embarked')
27 plt.xlabel('Embarked')
28 plt.ylabel('Count')
29 plt.legend(['Not Survived', 'Survived'])
```

Terminal Test cases

## 5.2.5. Bar Plot for Survival by Pclass

01:54

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value\_counts()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "**Survival by Pclass**" to the chart.
4. Label the x-axis as '**Pclass**' and the y-axis as '**Count**'.
5. The legend should indicate '**Not Survived**' and '**Survived**'.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

## Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
```

## BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15 # Write your code here for Bar Plot for Survival by Pclass
16
17 # Group by Pclass and Survived, then use value_counts()
18 survival_counts = data.groupby('Pclass')
19 ['Survived'].value_counts().unstack()
20
21 # Plot the stacked bar chart
22 survival_counts.plot(kind='bar', stacked=True)
23
24 # Add titles and labels
25 plt.title('Survival by Pclass')
26 plt.xlabel('Pclass')
27 plt.ylabel('Count')
28
29 # Add legend with custom labels
```

[Terminal](#) [Test cases](#)

## 5.2.4. Bar Plot for Survival by Gender

11:39

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the 'Sex' column, then use the `value_counts()` function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "Survival by Gender" to the chart.
4. Label the x-axis as 'Gender' and the y-axis as 'Count'.
5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

## Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
```

## BarPlotOf...

SUBMIT

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Gender
17 plt.figure()
18 data.groupby('Sex')
19 ['Survived'].value_counts().unstack().plot(kind='bar', stacked=True)
20 plt.title('Survival by Gender')
21 plt.xlabel('Gender')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()
25
```

[Terminal](#) [Test cases](#)[PREV](#) [RESET](#) [SUBMIT](#) [NEXT](#)

## 5.2.3. Bar plot of survival rate of passengers

11:01

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the **'Survived'** column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to **'bar'**.
3. Add the title **"Survival Count"** to the chart.
4. Label the x-axis as **'Survived'** and the y-axis as **'Count'**.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

## Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

## BarPlotOf...

SUBMIT

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival Rate
17 plt.figure()
18 data['Survived'].value_counts().plot(kind='bar')
19 plt.title('Survival Count')
20 plt.xlabel('Survived')
21 plt.ylabel('Count')
22 plt.show()
23
```

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## 5.2.2. Histogram of passenger information of Titanic

06:27

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**.
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

## Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

## Histogram...

SUBMIT

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Histogram
17 plt.figure()
18 plt.hist(data['Age'], bins=30, edgecolor='k')
19 plt.title('Age Distribution')
20 plt.xlabel('Age')
21 plt.ylabel('Frequency')
22 plt.show()
```

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## 5.2.1. Titanic Dataset

02:38

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

**Dataset Information:**

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

**Visualization:**

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

**Visualization Details:**

Write the code to create a series of visualizations as follows:

**Bar Plot (Pclass Distribution):**

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.
- Title the plot as **"Passenger Class Distribution"**.
- Label the x-axis as **"Pclass"** and the y-axis as **"Count"**.

**Pie Chart (Gender Distribution):**

- Create a pie chart to display the distribution of male and female passengers.

titanicDat...

SUBMIT

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset from the CSV file
5 df = pd.read_csv('titanic.csv')
6
7 # Set up the figure for 5 subplots
8 fig, axes = plt.subplots(3, 2, figsize=(12, 12))
9 # write the code..
10 # Plot 1: counts of passengers by class
11 axes[0, 0].bar(df['Pclass'].value_counts().index,
12               df['Pclass'].value_counts(), color='skyblue')
13 axes[0, 0].set_title("Passenger Class Distribution")
14 axes[0, 0].set_xlabel("Pclass")
15 axes[0, 0].set_ylabel("Count")
16
17 # Plot 2: Gender distribution
18 axes[0, 1].pie(df['Gender'].value_counts(),
19               labels=df['Gender'].value_counts().index,
20               autopct='%1.1f%%', colors=['lightblue', 'lightcoral'])
21 axes[0, 1].set_title("Gender Distribution")
22
23 # Plot 3: Age distribution
24 axes[1, 0].hist(df['Age'].dropna(), bins=8,
25               color='lightgreen', edgecolor='black')
26 axes[1, 0].set_title("Age Distribution")
27 axes[1, 0].set_xlabel("Age")
28 axes[1, 0].set_ylabel("Frequency")
29
```

Terminal Test cases

&lt; PREV RESET SUBMIT NEXT &gt;

## 5.1.1. Stacked Plot

03:07



Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases



stackedpl...

SUBMIT

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Data for Months and Temperature for three cities
5 data = {
6     'Month': ['January', 'February', 'March', 'April', 'May', 'June',
7              'July', 'August', 'September', 'October', 'November', 'December'],
8     'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
9     'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
10    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
11 }
12 df = pd.DataFrame(data)
13 plt.stackplot(df['Month'], df['City_A_Temperature'],
14              df['City_B_Temperature'], df['City_C_Temperature'])
15 plt.xlabel('Month')
16 plt.ylabel('Temperature')
17 plt.title('Temperature Variation')
18 plt.legend(loc='upper left')
19 plt.show()
20
21 # Write your code...
```

Terminal

Test cases

&lt; PREVIOUS

RESET

SUBMIT

NEXT &gt;